

MODULE 02

HTML Content & Media

Make Meaning Visible

A practical guide to structuring text, embedding media, building tables, and safely integrating external content.

Choose elements for the meaning

Before writing a tag, ask: “What does this content represent?” The browser uses that choice to build an accessibility tree.

`<p> / <div>`

Generic Content

Use `<p>` for text blocks; use `<div>` only as a nonsemantic container.

```
<div> <p>Welcome to the  
campus event.</p> </div>
```

`<section>`

Thematic Grouping

A thematic group that usually has a heading.

```
<section> <h2>Today's  
Schedule</h2> <p>Workshops  
at 9 AM.</p> </section>
```

`<article>`

Independent Content

A self-contained piece that stands on its own.

```
<article> <h2>New Library  
Hours</h2> <p>Open 24/7  
now.</p> </article>
```

`<aside>`

Tangential Content

Related material such as sidebars or call-outs.

```
<aside> <h3>Related  
Links</h3> <ul><li>  
<a>Campus Map</a></li></ul>  
</aside>
```

Images need source and purpose

The `<img>` tag is a void element (no closing tag). It requires attributes to locate the file and explain its meaning.

`src=""`

The path to the image file. If the path breaks, the image breaks.

`alt=""`

Alternative text for screen readers or if the image fails to load.

`width="" & height=""`

Reserves space on the page before the image loads, preventing layout shifts.

THE COMPLETE ELEMENT

```

```

MEANINGFUL IMAGE (REQUIRES ALT TEXT)

```
<!-- The image conveys data or context -->

```

DECORATIVE IMAGE (EMPTY ALT)

```
<!-- The image is purely for visual flair -->

```

Alt text communicates the image job

Alternative text replaces the image when it cannot be seen. It should convey the **purpose** of the image, not just describe its pixels.

PRO TIP

Never write "Image of..." or "Picture of..."

Screen readers already announce that it is an image. Redundant text wastes the user's time.

DOES THE IMAGE CONTAIN TEXT?

Use that exact text.

```
alt="Campus Connect 2026"
```



DOES IT CONVEY MEANING OR DATA?

Describe the meaning or function.

```
alt="Map showing library north of science building"
```



IS IT PURELY DECORATIVE?

Use an empty alt attribute.

```
alt=""
```

Responsive images serve the right resource

Don't send a massive desktop image to a small mobile phone. HTML provides tools to let the browser choose the best image for the current screen.

RESOLUTION SWITCHING

Provide multiple sizes of the **same image**. The browser calculates the screen size and pixel density, then downloads only the smallest file it needs.

ART DIRECTION

Provide **different images** (e.g., cropped differently) for different screen sizes. You force the browser to use a specific image at a specific breakpoint.

RESOLUTION SWITCHING (SRCSET & SIZES)

```

```

ART DIRECTION (<PICTURE>)

```
<picture>
  <source media="(min-width: 800px)" srcset="hero-
wide.jpg">
  
</picture>
```

Figures keep media and context together

When media needs an explanatory caption, wrap it using `<figure>` and `<figcaption>` so the caption stays tied to its media.

THE SEMANTIC CONNECTION

Wrapping an image and its caption in a `<figure>` makes the relationship explicit to assistive technologies.

NOT JUST FOR IMAGES

Figures work for charts, code blocks, or any referenced media that could be moved without breaking the flow.

```
<figure>
  
  <figcaption>
    Fig 1. The new
    library is located north
    of the science building.
  </figcaption>
</figure>
```

BROWSER RENDER

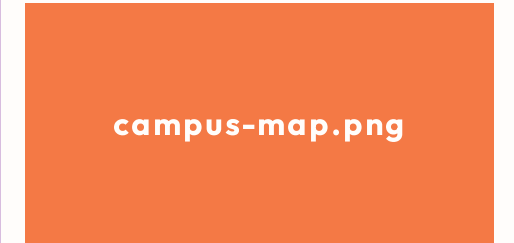


Fig 1. The new library is located north of the science building.

Audio and video need real fallbacks

Native HTML5 media lets you embed audio and video without plugins. Accessibility and fallbacks are mandatory.

■ Provide Controls

Include the `controls` attribute so users can control playback.

■ Include Captions

Add a `<track>` (VTT) for subtitles so deaf users can follow along.

■ Write Fallback Content

Provide brief text or a download link when media can't play.

VIDEO WITH CAPTIONS AND FALLBACK

```
<video controls width="640" height="360">
  <source src="lecture.mp4" type="video/mp4">
  <source src="lecture.webm" type="video/webm">
  <track kind="captions" src="lecture-en.vtt"
        srclang="en" label="English" default>
  Your browser doesn't support HTML video. Download.
</video>
```

AUDIO WITH FALLBACK

```
<audio controls>
  <source src="podcast-ep1.mp3" type="audio/mpeg">
  Your browser doesn't support HTML audio. Download audio.
</audio>
```

Tables represent two-dimensional data

HTML tables are designed for structured data with rows and columns. To make them accessible, define their semantic parts.

STRICT RULE

Never use tables for page layout. Use CSS Grid or Flexbox instead.

<caption>

The title of the table. Important for screen readers.

<tr>

Defines a table row.

<td>

Standard data cell.

<thead> & <tbody>

Groups header rows and data rows.

<th scope="...">

Header cell; use scope to link to data.

```
<table>
  <caption>Workshop Schedule</caption>
  <thead>
    <tr>
      <th scope="col">Time</th>
      <th scope="col">Topic</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">09:00</th>
      <td>HTML Basics</td>
    </tr>
  </tbody>
</table>
```

RENDERED OUTPUT (WITH CSS)

Workshop Schedule

Time	Topic
09:00	HTML Basics
11:00	CSS Layouts

Progressive disclosure reduces noise

Hide secondary content behind a toggle using the native `<details>` element.

Zero JavaScript Required

The browser handles open/close natively — no JS needed.

Native Keyboard Support

Summary can be focused and toggled with keyboard keys.

```
<details> <summary>View prerequisites</summary>
<p>Students should know how to use a browser and editor.
</p> </details>
```

BROWSER RENDER (CLOSED STATE)

▶ **View prerequisites**

BROWSER RENDER (OPEN STATE)

▼ **View prerequisites**

Students should know how to use a browser and editor.

Embeds require trust boundaries

The `<iframe>` element embeds another HTML page into yours. Because you are loading external code, you must establish strict security and performance boundaries.

title=""

Essential for screen readers to announce what the frame contains before the user enters it.

sandbox=""

Restricts what the embedded content can do (e.g., blocks scripts or popups) unless explicitly allowed.

loading="lazy"

Defers loading the iframe until it reaches the viewport, saving bandwidth and speeding up your page.

A SECURE IFRAME EMBED

```
<iframe
  src="https://maps.example.com/embed/campus"
  title="Interactive map of the main campus"
  width="600" height="400"
  loading="lazy"
  referrerpolicy="no-referrer"
  <!-- Only allow scripts, block popups/forms -->
  sandbox="allow-scripts">
</iframe>
```

WHEN IS A LINK BETTER?

Third-party embeds (like social media posts or maps) often load megabytes of tracking scripts. If the embed isn't essential to the immediate context, a simple `<a href="...">` is safer, faster, and more respectful of user privacy.

Build a rich event page

Bringing it all together. A semantic document structure enriched with accessible media, tabular data, and progressive disclosure.

```
<body>
  <header>    <h1>Campus Connect</h1></header>
  <main>
    <article>
      <h2>Tech Career Fair 2026</h2>
      <figure>
<figcaption>Main Exhibition Hall</figcaption></figure>
      <table><caption>Speaker Schedule</caption><!-- thead, tbody,
tr... --></table>
      <details><summary>Parking Information</summary><p>Use the
North Lot (Lot B).</p></details>
    </article>
  </main>
</body>
```

SEMANTIC WIREFRAME

<header>

<main>

<article>

<figure> + + <figcaption>

<table>

<th>

<th>

<td>

<td>

<details> & <summary>

Content lab: information with context

DELIVERABLES

- 1 Build an **<article>** for a Campus Connect event, including a thematic **<section>** for the schedule.
- 2 Add a responsive **<picture>** hero image that serves a different crop on small screens.
- 3 Create a semantic **<table>** for the event itinerary, complete with a caption and proper header scopes.
- 4 Use **<details>** and **<summary>** to hide optional parking and transit information.

VALIDATION CHECKS

- ✓ Are all alt attributes purposeful (or intentionally empty if decorative)?
- ✓ Can you navigate to and toggle the **<details>** element using only your keyboard?
- ✓ Does the table use **<th scope="col">** for its column headers?

EXTENSION CHALLENGE

Embed a campus map using an **<iframe>**. You must apply a strict **sandbox** attribute, provide a descriptive **title**, and ensure it uses **loading="lazy"** to protect page performance.